ICCV OCT 19-23, 2025 HONOLULU HAWAII

Foundation Models in Visual Anomaly Detection: Advances, Challenges, and Applications

Organizers: Jiawen Zhu, Peng Wu, Olga Fink, Chengjie Wang, Guansong Pang

Guest Speakers: Ismail Nejjar, Jinlong Peng

ICCV 2025 Tutorial

Oct. 19, 2025, 9:00 AM - 12:00 PM (PDT)

SMU SINGAPORE MANAGEMENT UNIVERSITY

西北工業大學 NORTHWESTERN POLYTECHNICAL UNIVERSITY

EPFL



Anomaly Generation with Foundation Models

Ismail Nejar

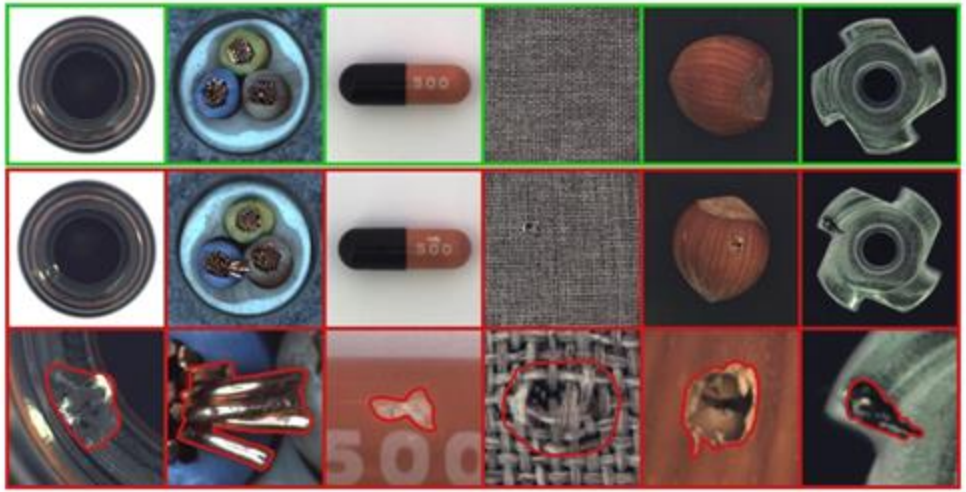
Intelligent Maintenance and Operations Systems (IMOS)
EPFL



Automated Industrial Inspection

Industrial Inspection & Quality Control

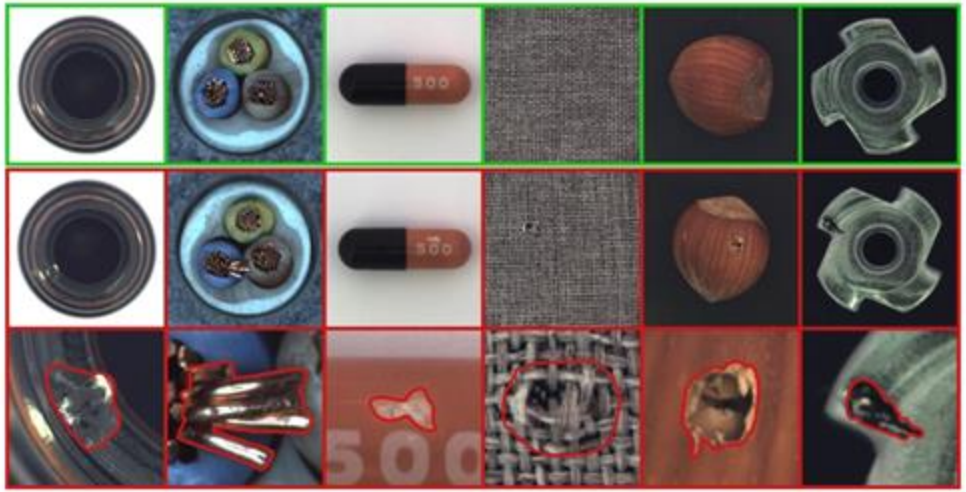
- Detecting unusual patterns in images



Challenge

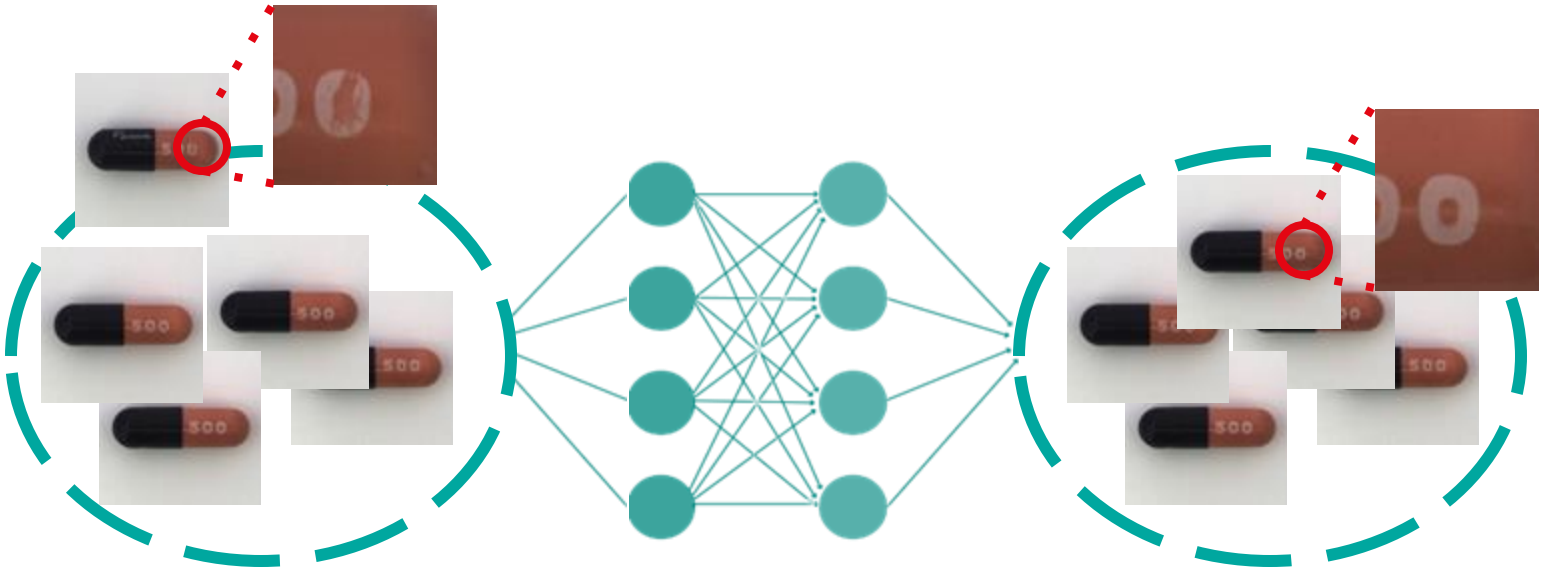
We might not know how the anomalies looks like

- Rare faulty collection
- High cost to obtain labeled data



Traditional Visual anomaly detection

Learning the distribution of normal samples without any anomalies



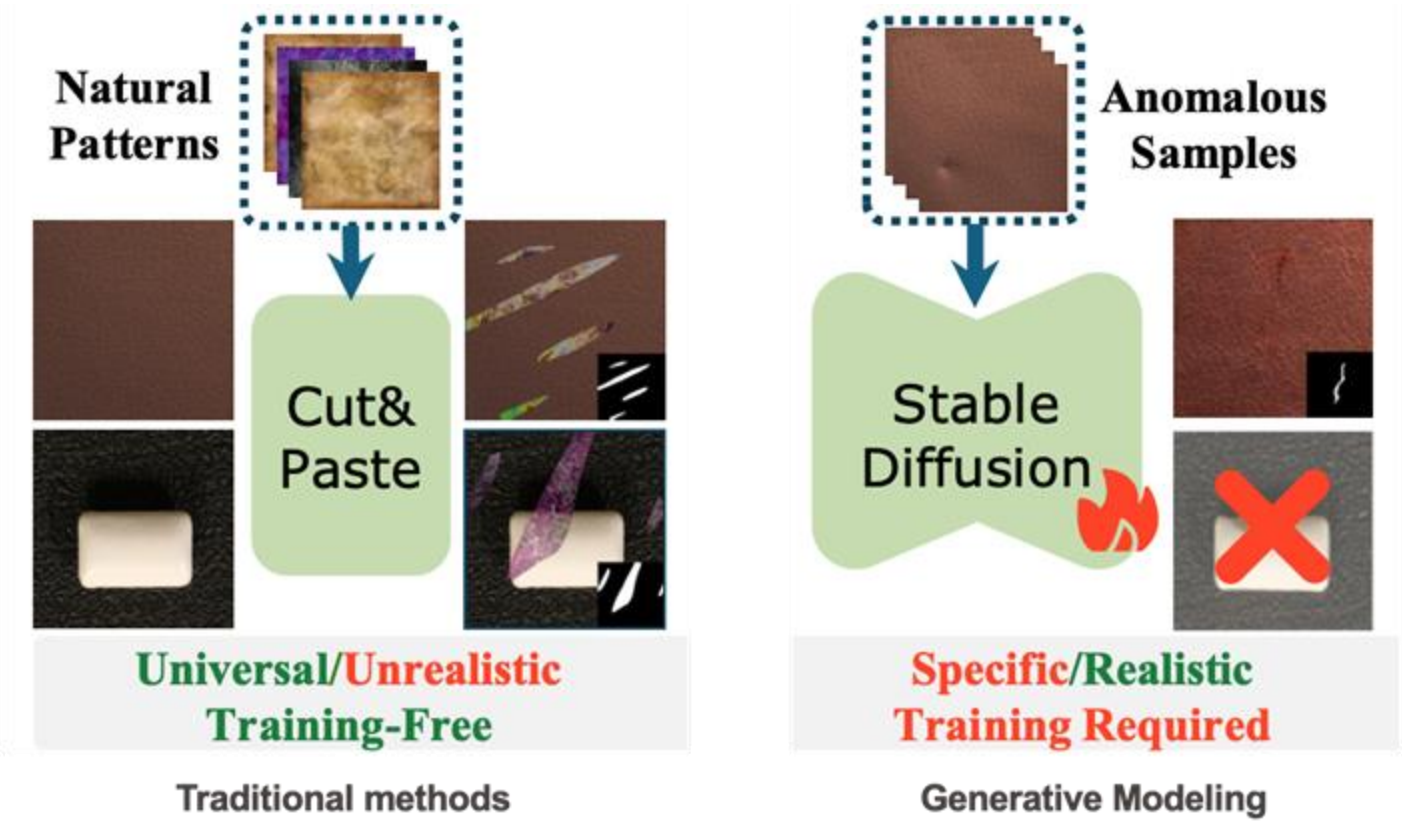
Why anomaly generation ?

HOWEVER, we might neither know how the normal samples look like

- Variety in products
- Variety of applications
- New product

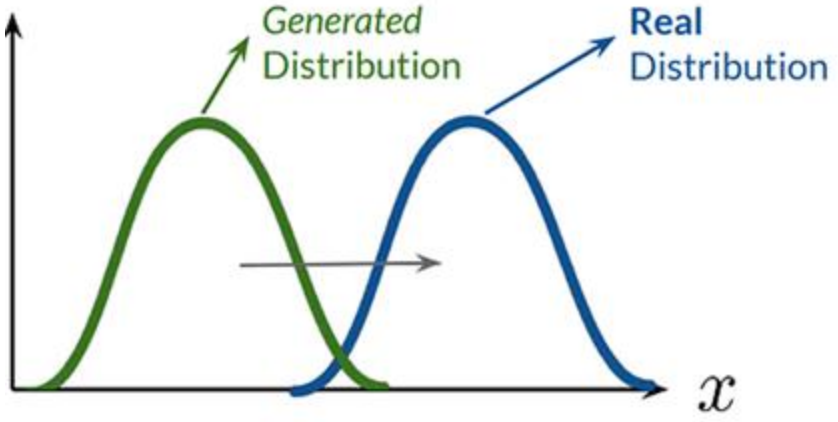
Let’s generate more anomalous samples!

Visual anomaly generation



Generative Modeling

Goal: Learning the underlying distribution of the data





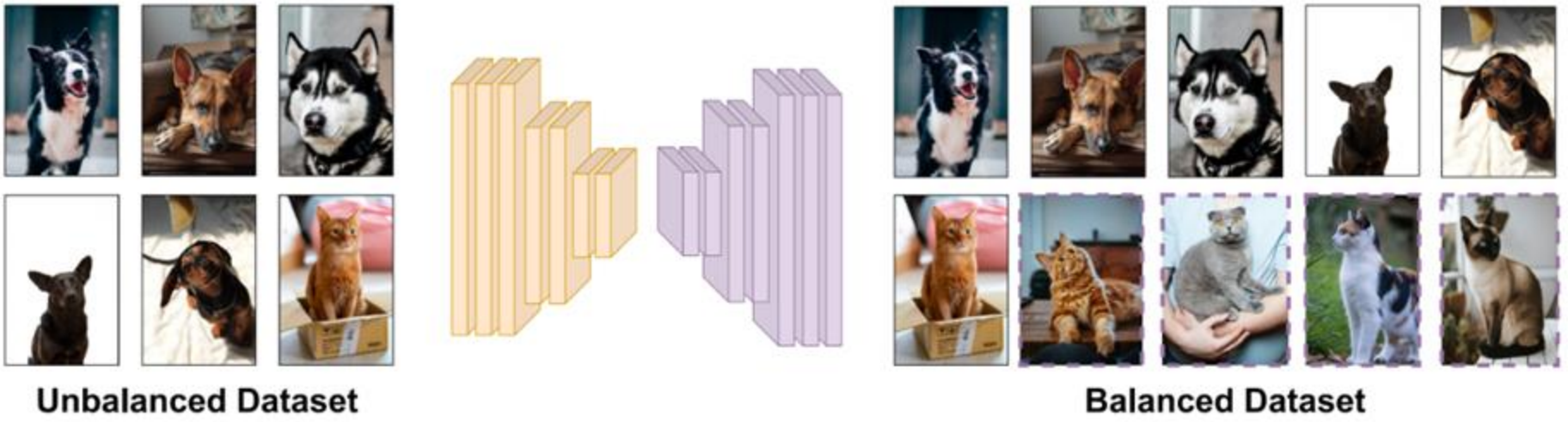
Input ~ $P_{data}(x)$



Output~$P_{model}(x)$

Crafting the Unknown with Generative Models

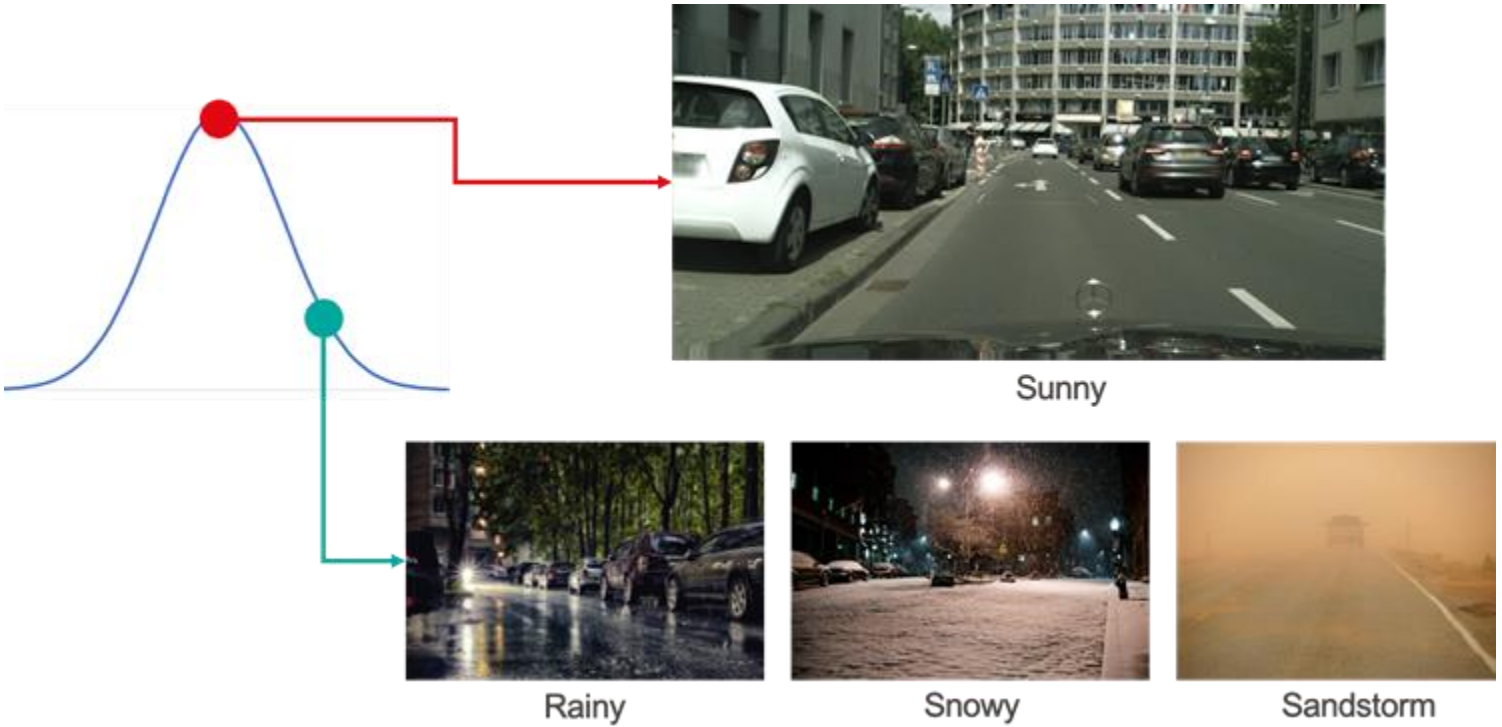
Debiasing



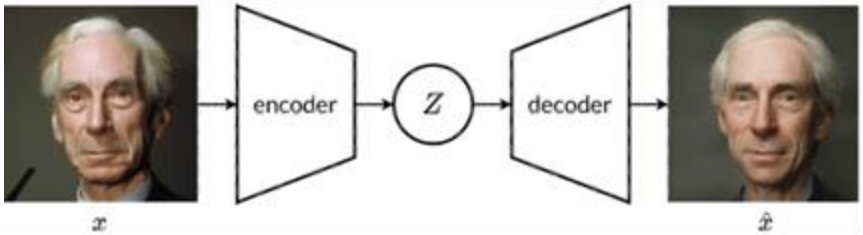
Parraga, Otávio, et al. "Debiasing methods for fairer neural models in vision and language research: A survey." *arXiv preprint arXiv:2211.05617* (2022).

Crafting the Unknown with Generative Models

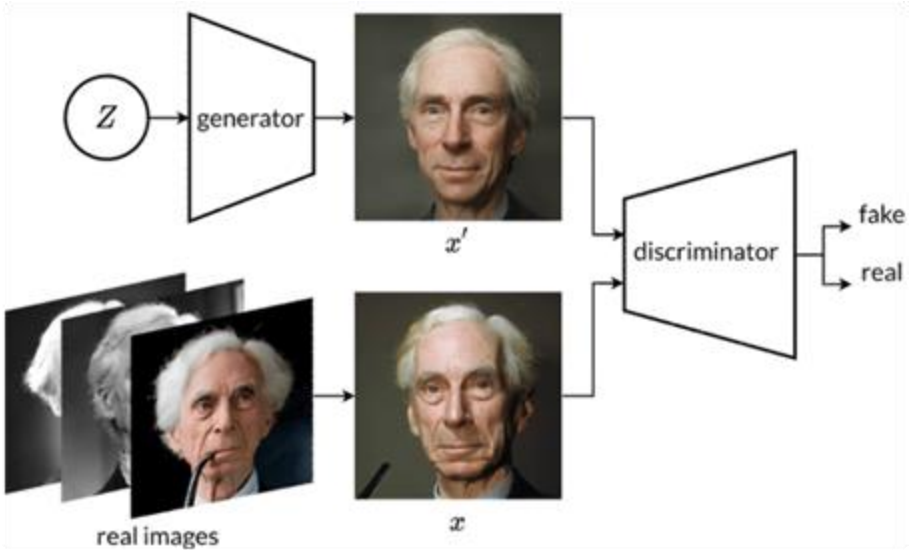
Generalizing



How does generative modeling work ? (1)



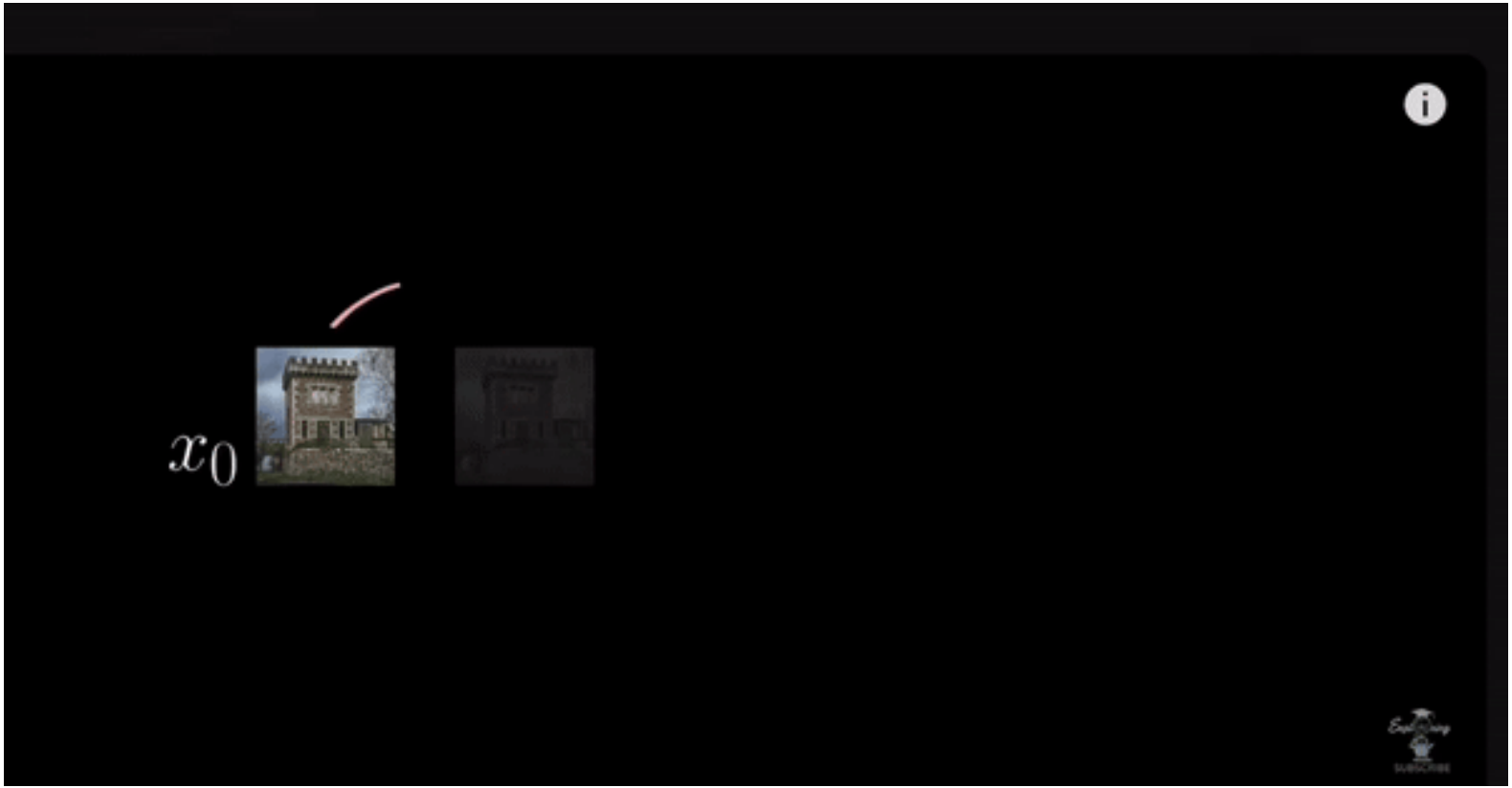
Autoencoder (AE)



Generative Adversarial Networks (GAN)

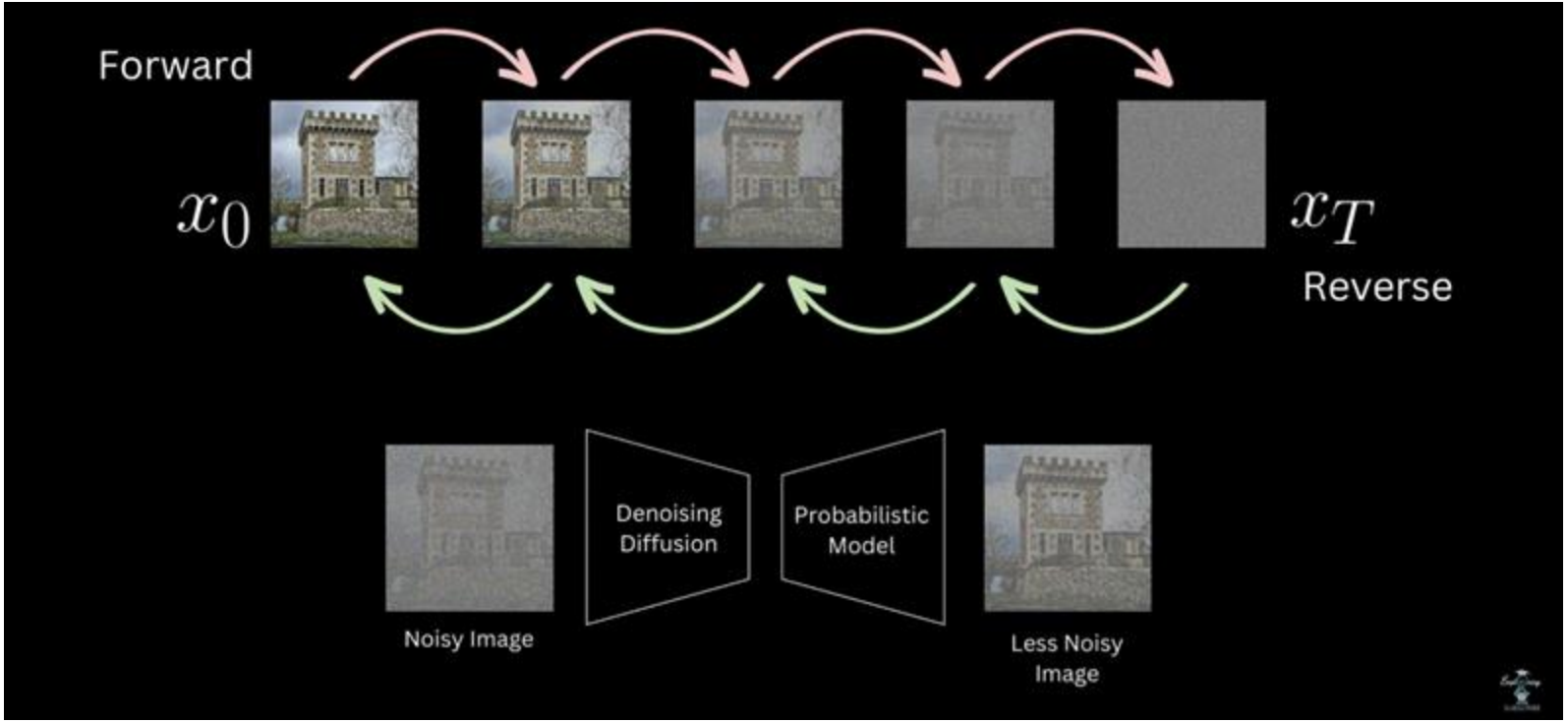
https://vitalflux.com/gan-vs-vae-differences-similarities-examples/

How does generative modeling work ? (2)



https://www.youtube.com/watch?v=H45lF4sUgiE&list=PL8VDJoEXIjpo2S7X-1YKZnbHyLGyESDCe

How does generative modeling work ? (2)



https://www.youtube.com/watch?v=H45lF4sUgiE&list=PL8VDJoEXIjpo2S7X-1YKZnbHyLGyESDCe

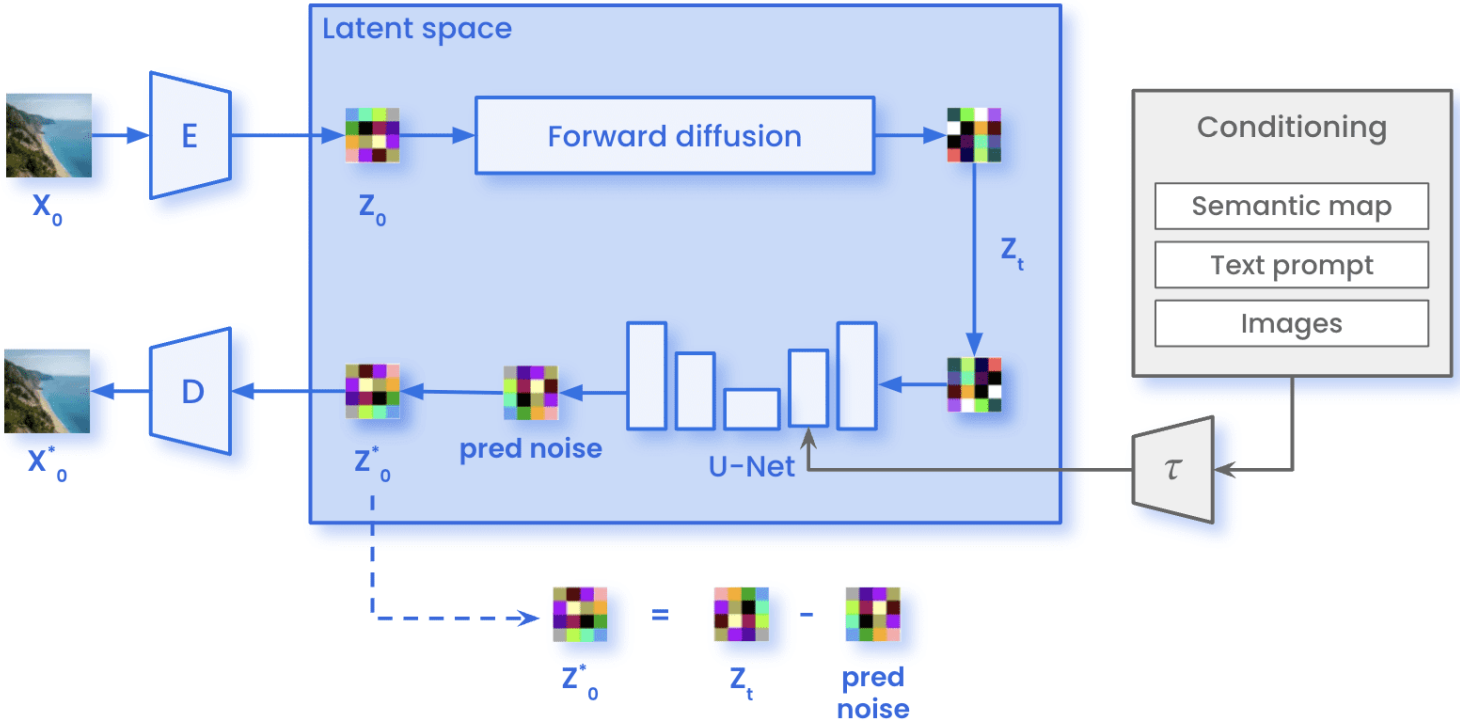
Stable Diffusion

Stable diffusion is a latent text-to-image diffusion model:

- Pretrained on large scale dataset
- Realistic and diverse generation results



How Stable Diffusion Works ?



Stable Diffusion for anomalies

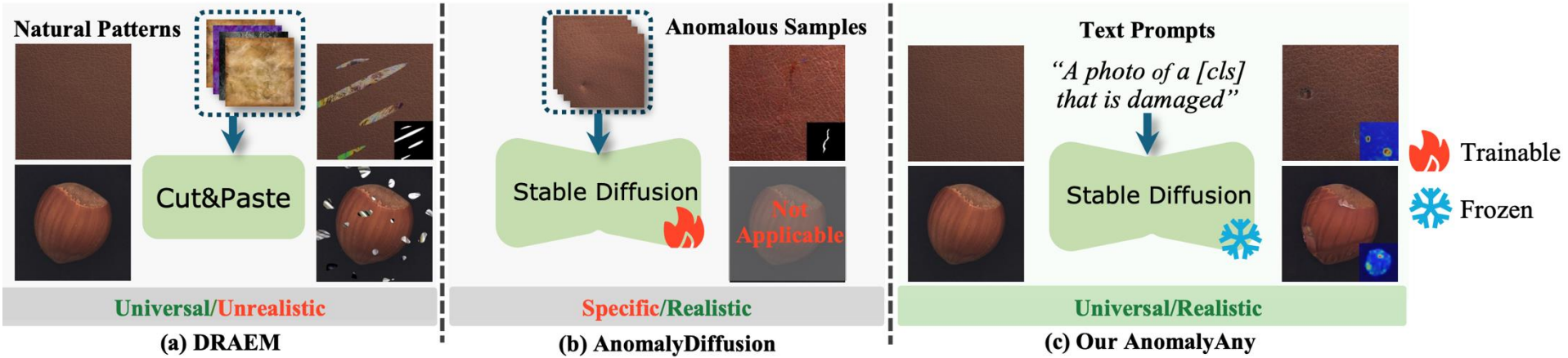
Anomalies are rare concepts in SD:

- Produce images that deviate from the desired normal distribution
- Fail to accurately represent realistic anomaly patterns



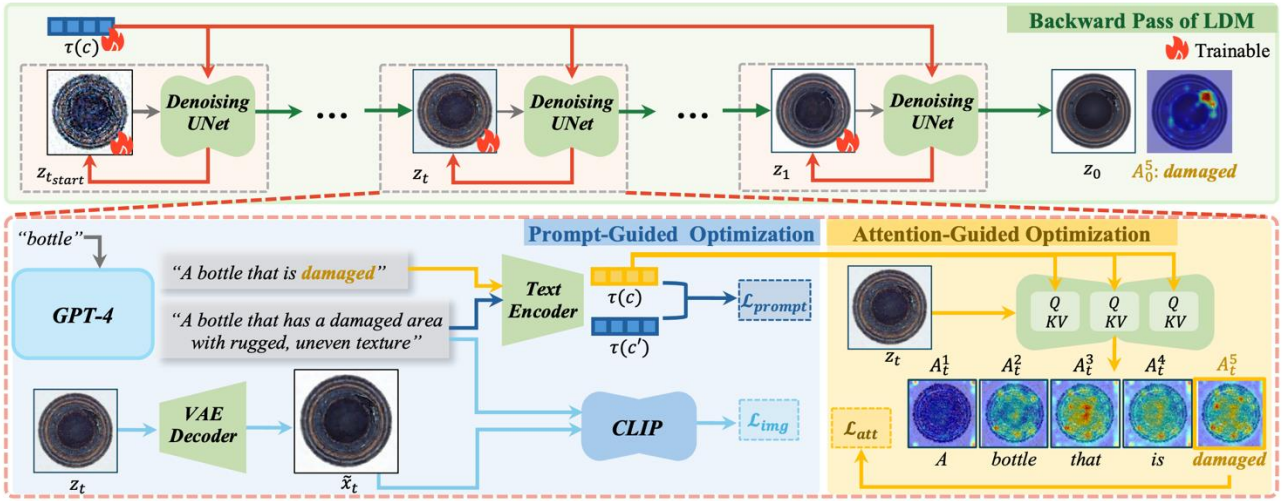
Proposed method: AnomalyAnything

- One-shot (normal sample + text description)
- No need for training
- Unseen anomaly generation on arbitrary anomalies

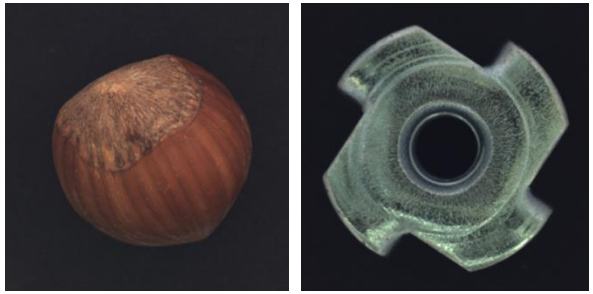


AnomalyAny Modules

- Test-time Normal Sample Conditioning
- Attention-Guided Anomaly Optimization
- Refined Anomaly Generation via Prompt Guided Optimization



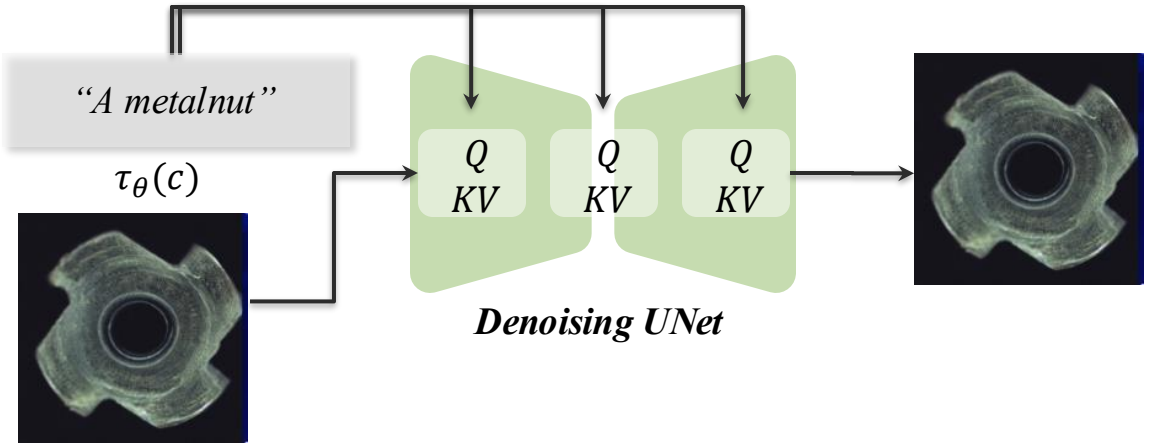
A hazelnut/ metal nut



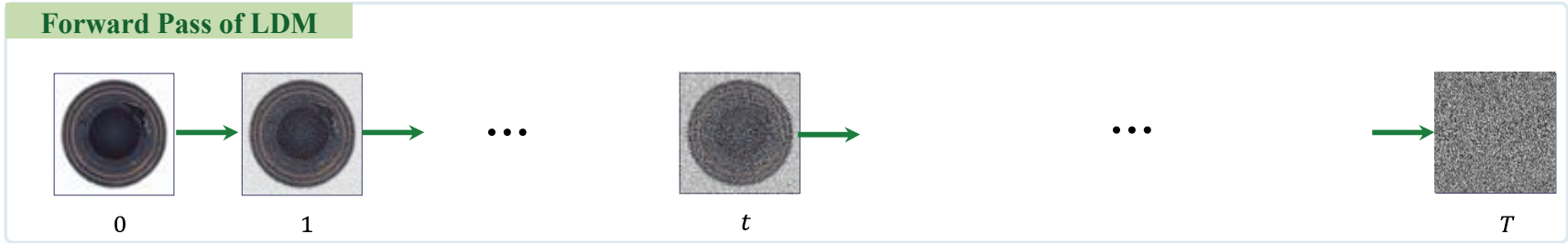
Desired Output

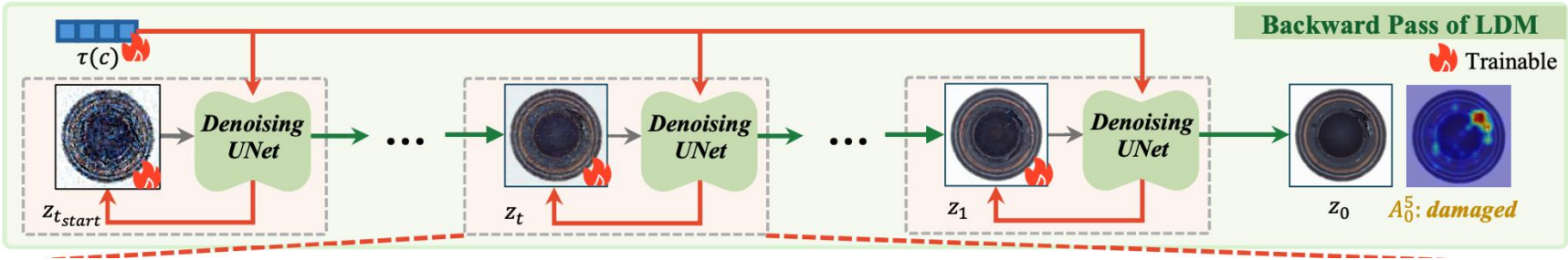


Generated Output

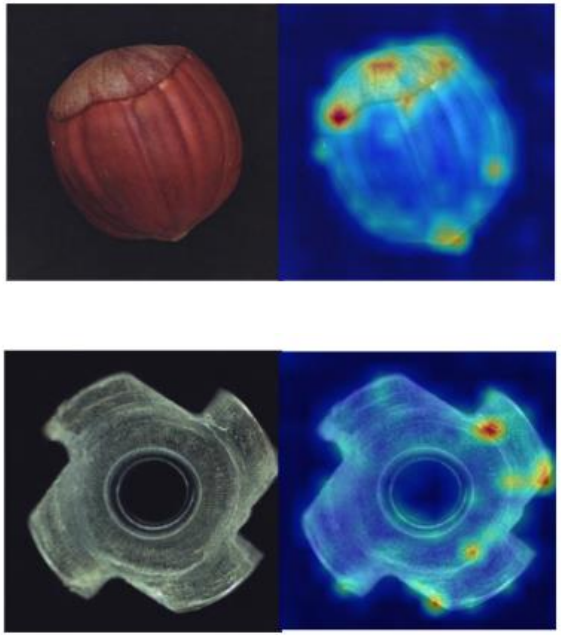


Test-time Normal Sample Conditioning



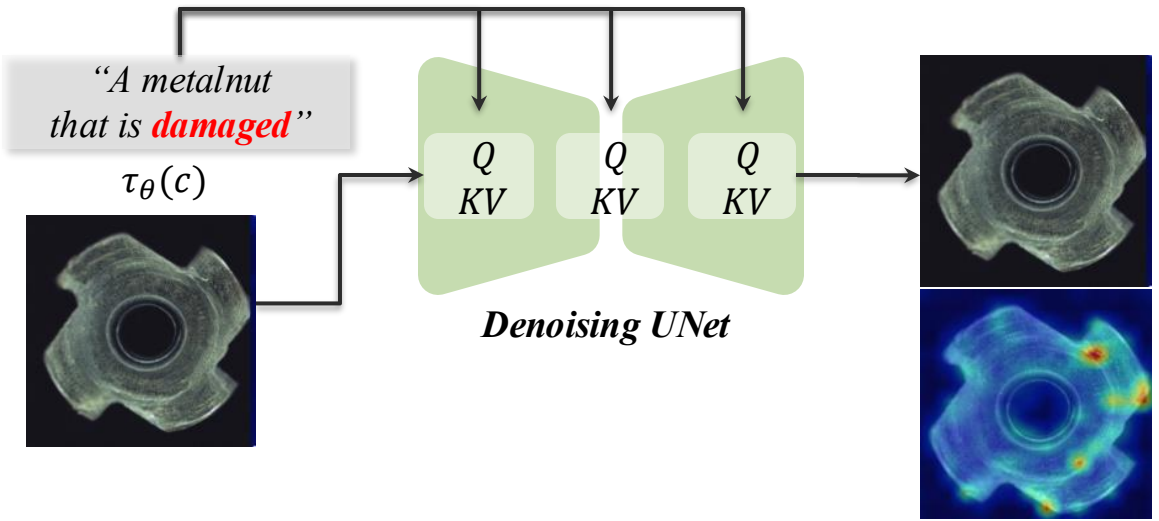


A hazelnut/ metal nut

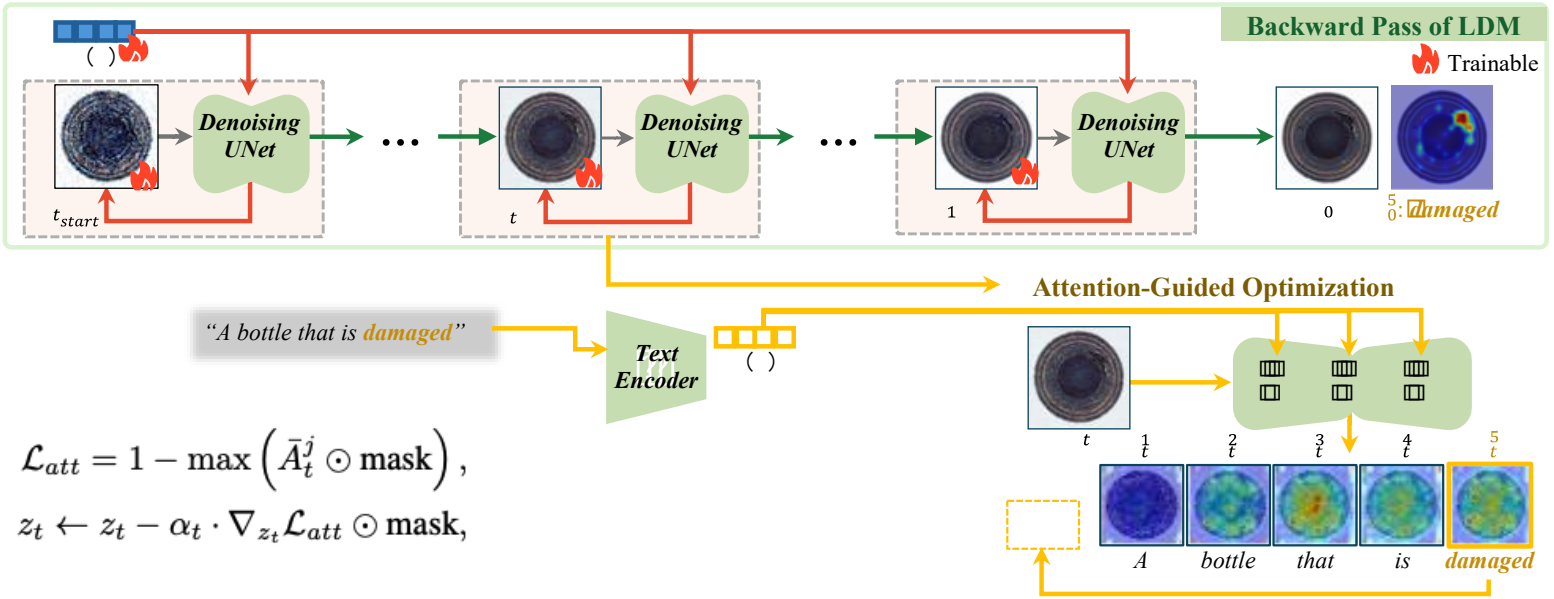


Generated Output

Anomaly is a hard concept for diffusion models.



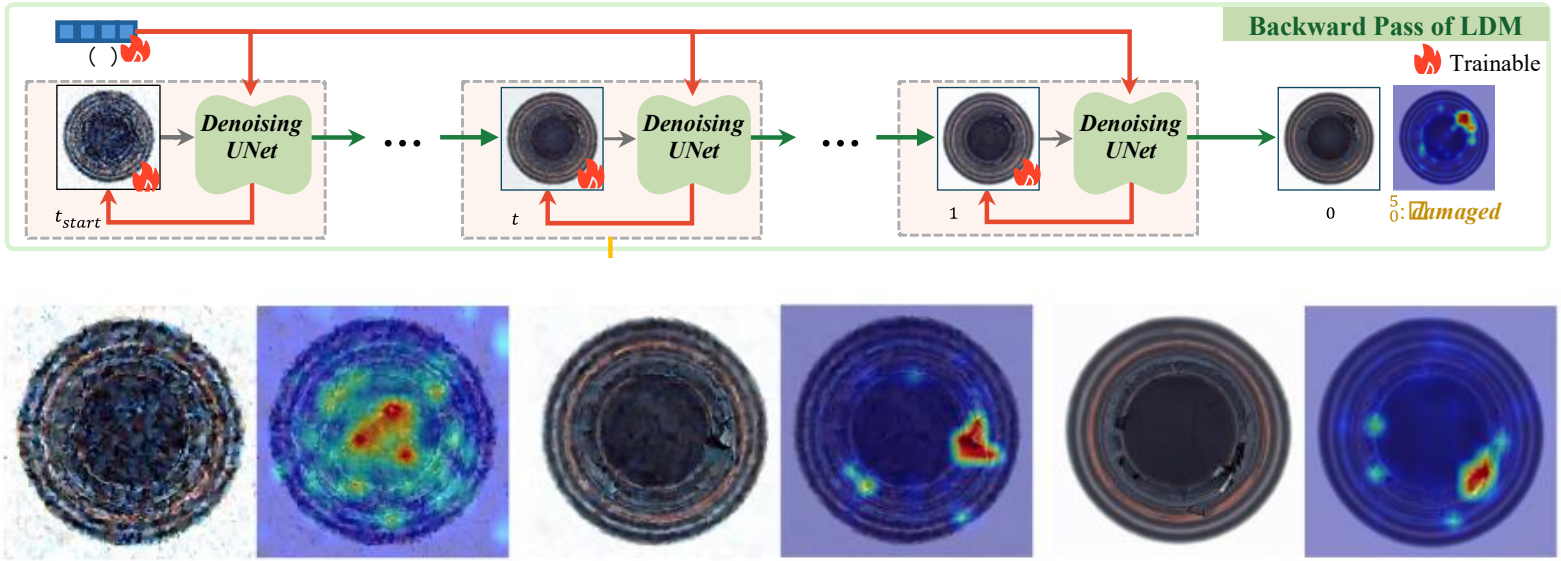
Attention guided anomaly optimization



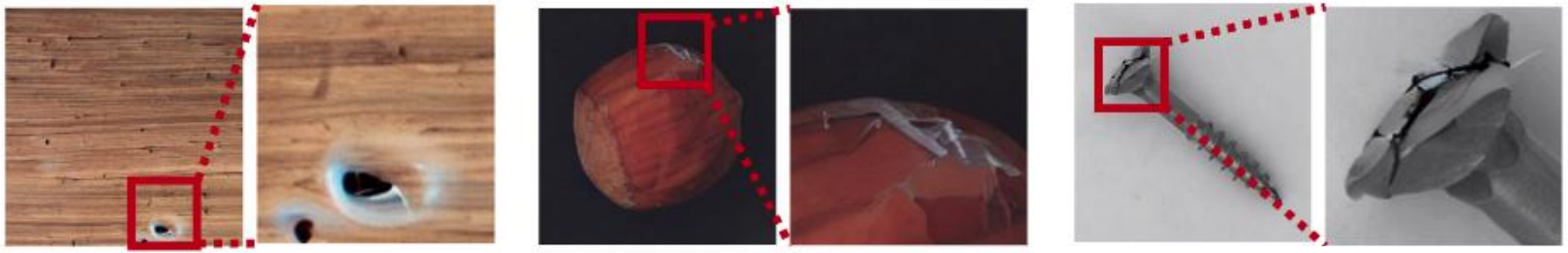
$$\mathcal{L}_{att} = 1 - \max\left(\bar{A}_t^j \odot \text{mask}\right),$$

$$z_t \leftarrow z_t - \alpha_t \cdot \nabla_{z_t} \mathcal{L}_{att} \odot \text{mask},$$

Attention guided anomaly optimization (2)

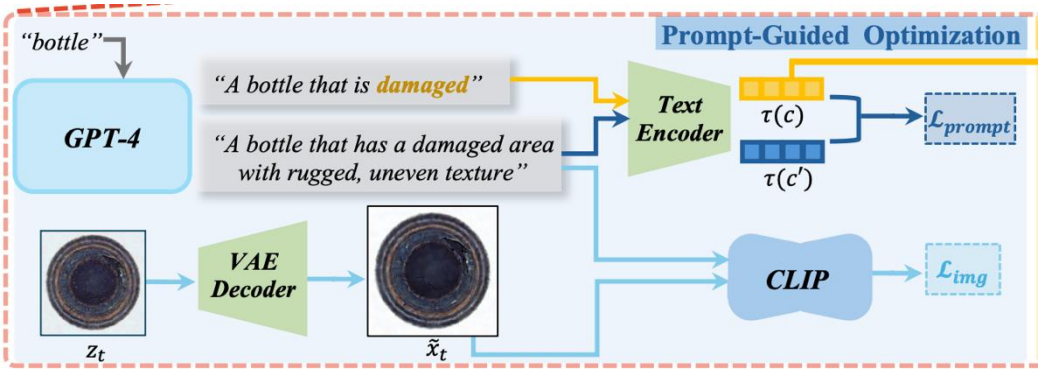


Attention guided anomaly optimization - Result



Over-optimization & artifacts

Refined anomaly generation via prompt-guided optimization



Latent optimization

$$\mathcal{L}_{img} = 1.0 - \text{cosine}(\Phi^T(c'), \Phi^V(\tilde{x}_t)),$$

$$\mathcal{L} = \mathcal{L}_{img} + \alpha_t \cdot \mathcal{L}_{att}, \quad z_t \leftarrow z_t - \nabla_{z_t}\mathcal{L} \odot mask.$$

Prompt optimization

$$\mathcal{L}_{prompt} = 1.0 - \text{cosine}(\tau(c), \tau(c')),$$

$$\tau(c) \leftarrow \tau(c) - \nabla_{\tau(c)}(\mathcal{L}_{prompt} + \mathcal{L}_{img}).$$

Refined anomaly generation via prompt-guided optimization – Result

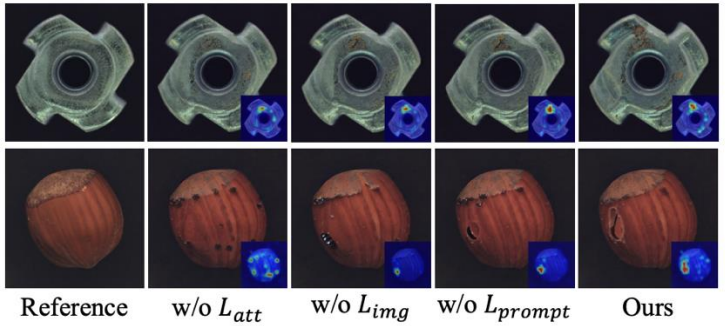
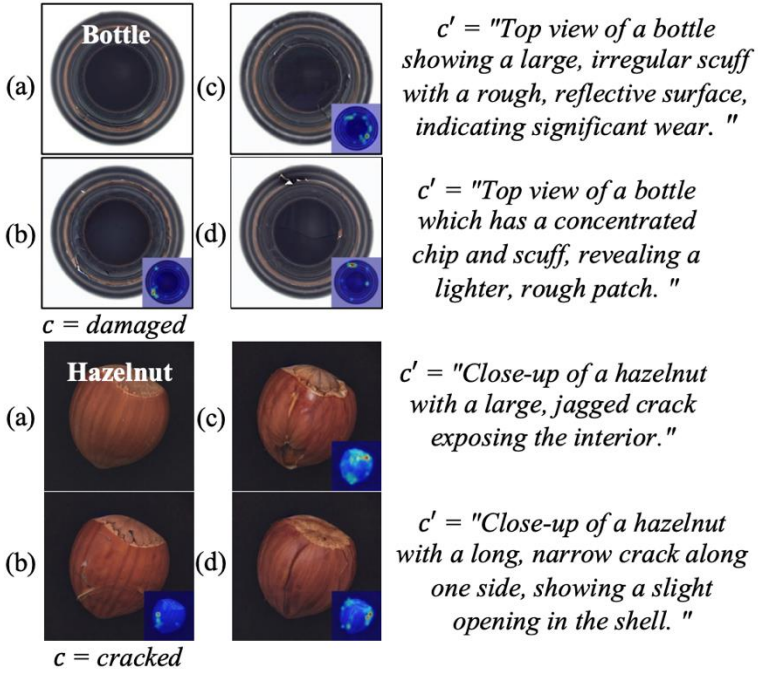
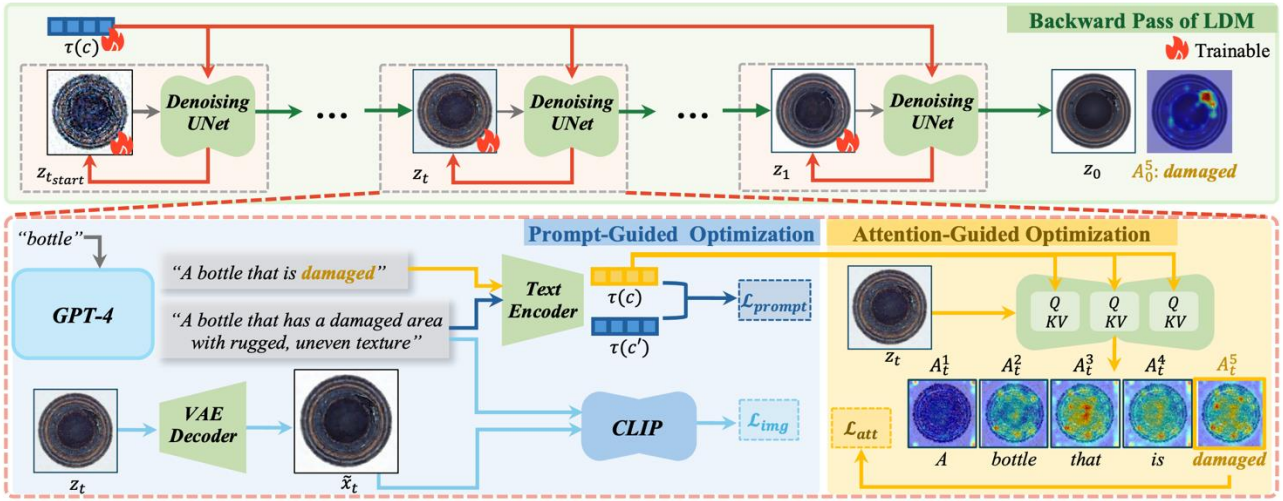


Figure 10. **Ablation on prompt-guided optimization objectives.** Anomaly types are *"rust"* and *"hole"* respectively.

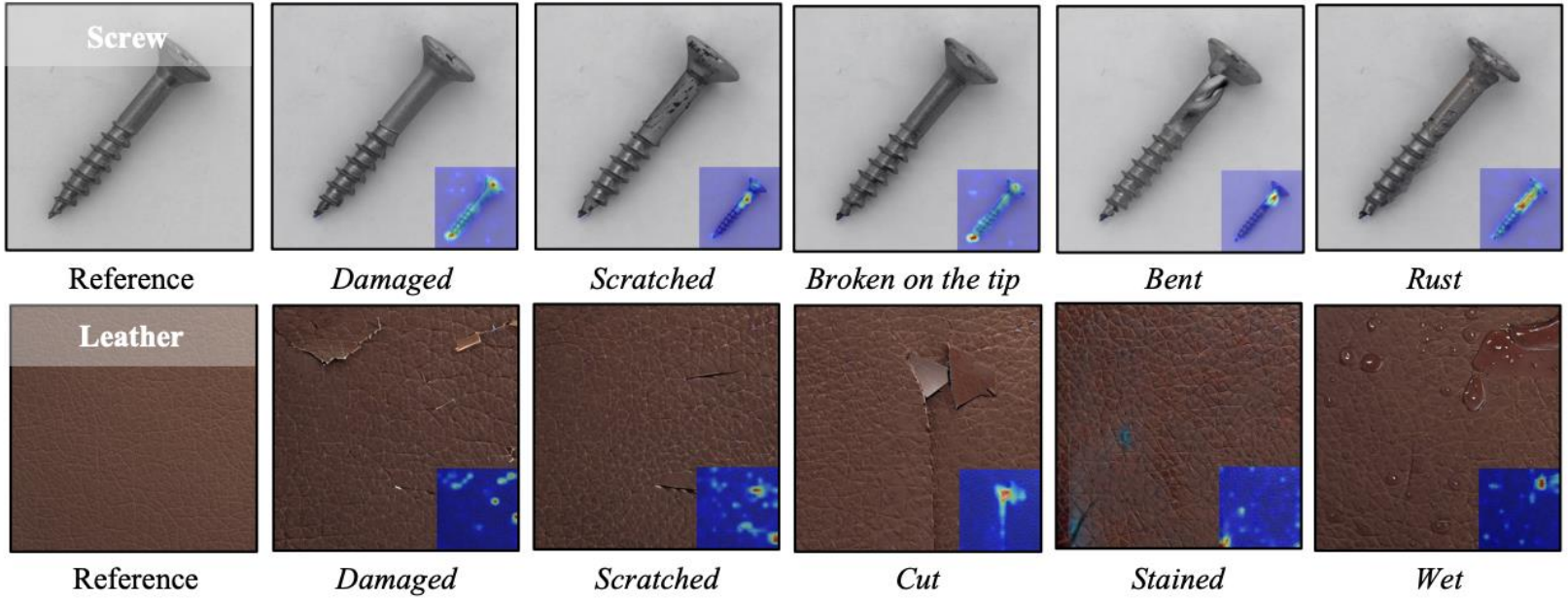


AnomalyAnything

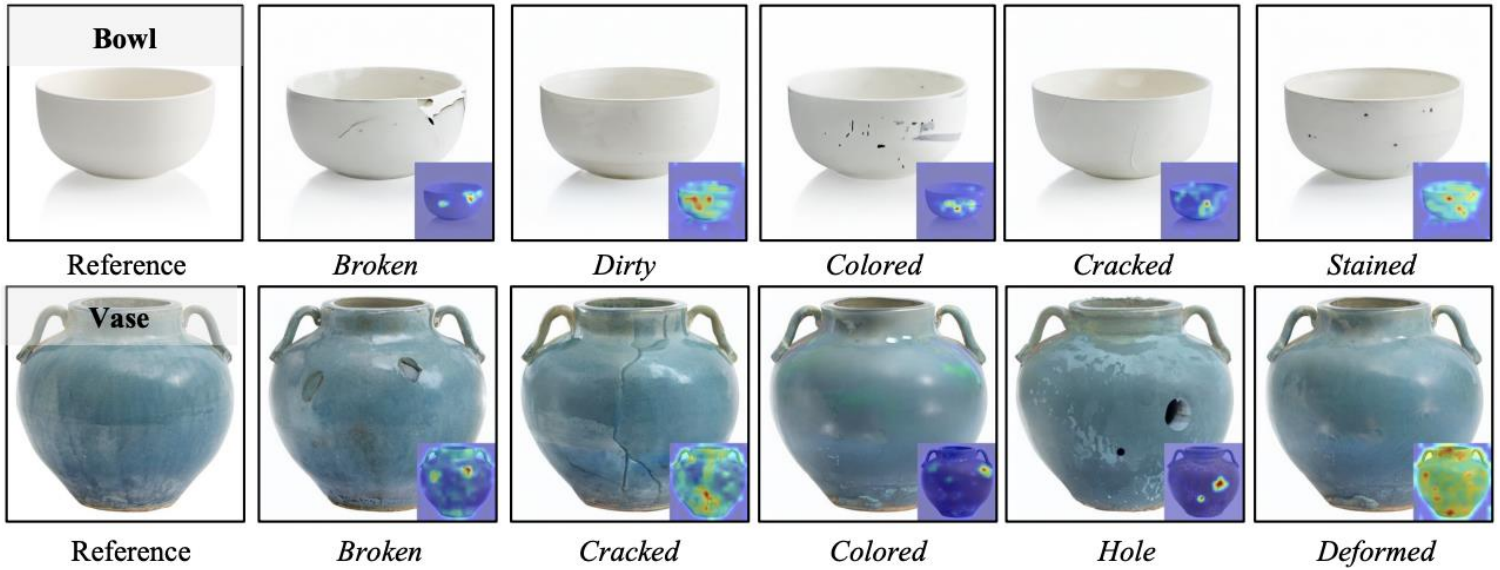
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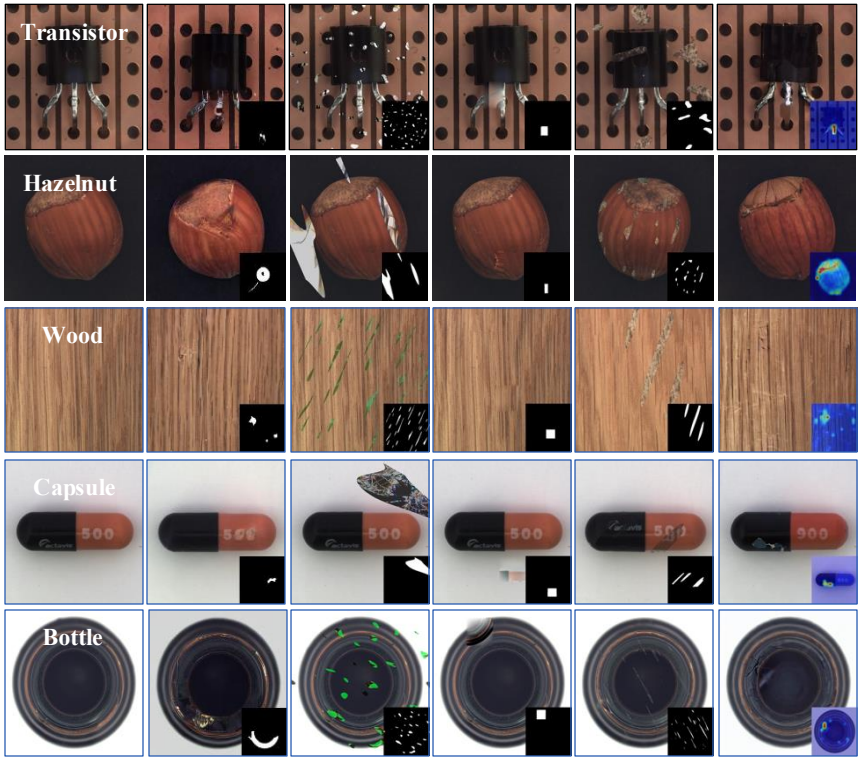
Generation with desired object and anomaly



Generation with desired object and anomaly



In comparison with other anomaly generation methods



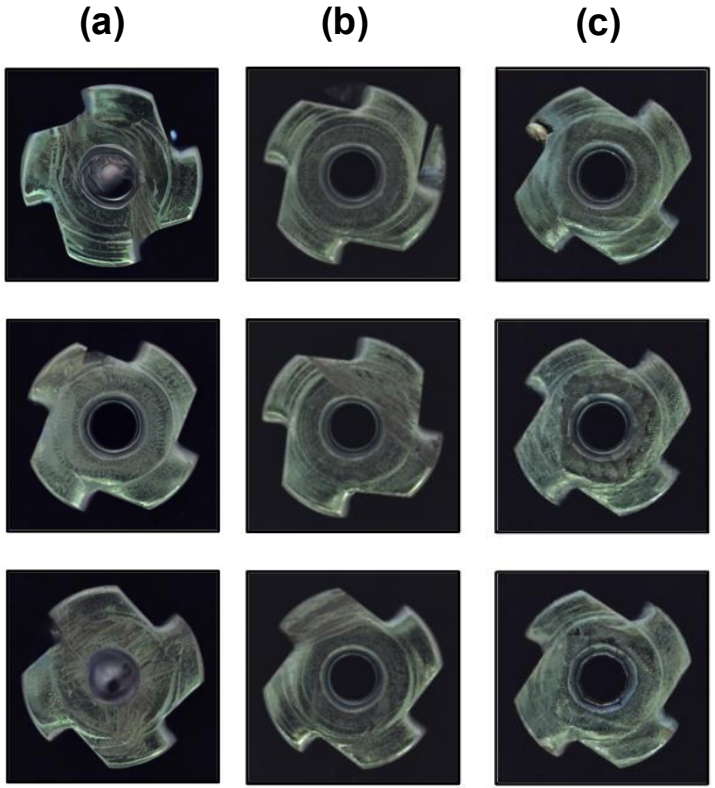
Category	NSA [29]		RealNet [35]		AnoDiff [17]		Ours	
	IS ↑	IC-L ↑	IS ↑	IC-L ↑	IS ↑	IC-L ↑	IS ↑	IC-L ↑
bottle	1.35	0.11	1.58	0.16	1.58	0.19	1.73	0.17
cable	1.55	0.38	1.65	0.41	2.13	0.41	2.06	0.41
capsule	1.38	0.17	1.62	0.18	1.59	0.21	2.16	0.23
carpet	1.03	0.22	1.03	0.25	1.16	0.24	1.10	0.34
grid	2.26	0.36	2.24	0.37	2.04	0.44	2.31	0.38
hazelnut	2.04	0.31	2.21	0.31	2.13	0.31	2.55	0.32
leather	1.07	0.26	1.67	0.36	1.94	0.41	2.26	0.41
metal nut	1.66	0.23	1.67	0.24	1.96	0.30	1.82	0.27
pill	1.34	0.24	1.45	0.26	1.61	0.26	2.91	0.30
screw	1.20	0.31	1.20	0.31	1.28	0.30	1.33	0.32
tile	1.31	0.37	1.57	0.48	2.54	0.55	2.66	0.53
toothbrush	1.18	0.17	1.19	0.18	1.68	0.21	1.64	0.22
transistor	1.34	0.22	1.49	0.30	1.57	0.34	1.66	0.28
wood	1.30	0.32	2.22	0.41	2.33	0.37	1.93	0.41
zipper	1.52	0.22	1.88	0.24	1.39	0.25	2.14	0.33
average	1.44	0.26	1.64	0.30	1.80	0.32	**2.02**	**0.33**

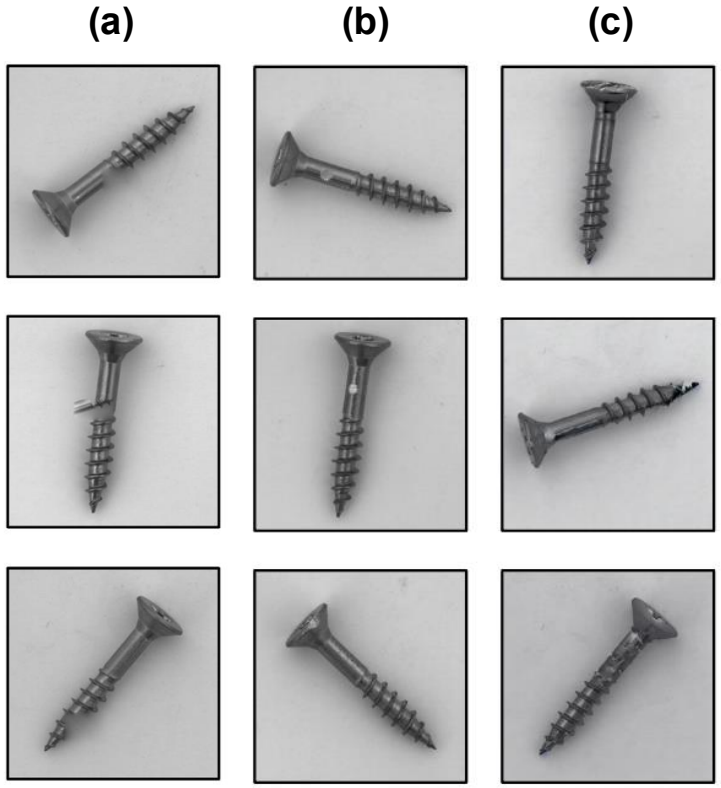
Table 1. **Comparisons between anomaly generation quality and diversity with different anomaly generation methods.** AnoDiff: AnomalyDiffusion.



Which one is real

Which is real?





Unseen Visual Anomaly Generation

Han Sun
EPFL, Switzerland

Yunkang Cao
HUST, China

Hao Dong
ETH Zurich, Switzerland

Olga Fink
EPFL, Switzerland

